

FPGA-Based VGG-16 CNN Hardware Accelerator with DDR3 Interface

4-Channel Parallel RTL Architecture on Xilinx Virtex-7 VC707

A fully parameterized, RTL-based hardware accelerator that executes the complete 13-layer VGG-16 CNN end-to-end on FPGA — achieving **83.8 GOP/s** at just **10.71% DSP utilization**, using a novel 4-engine parallel architecture with hierarchical FSM control and DDR3-aware dataflow scheduling.

HDL:	Verilog
Target FPGA:	Xilinx Virtex-7 VC707 (xc7vx485tffg1761-2)
Toolchain:	Xilinx Vivado 2016.4 (licensed)
Simulation Status:	✓ 100% verified, Blocks 1–5, full VGG-16

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July 4, 2026

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Tool & License Requirements

This project was fully developed, synthesized, and simulated in **Xilinx Vivado 2016.4**, targeting the **Virtex-7 VC707 Evaluation Platform (xc7vx485tffg1761-2)**.

Note: Vivado 2016.4 is a **licensed** tool — a valid Xilinx/AMD license is required to open, synthesize, or simulate this project. The VC707 board files and MIG (Memory Interface Generator) DDR3 core used in this design are also version- and license-dependent on the installed Vivado version.

To port this project to a newer version of Vivado, the following steps are required:

Step	Action
1	Obtain a valid Vivado license for the target version
2	Re-generate the MIG 7-Series DDR3 IP core (<code>mig_7series_0</code>) for compatibility
3	Import VC707 board files into the newer Vivado installation
4	Re-validate the IP Integrator block design (<code>design_1.bd</code>), since IP versions shift across Vivado releases

Once these are in place, the RTL itself (pure Verilog, technology-agnostic) requires **no logic changes** to re-target a newer toolchain.

Simulation Status

The complete VGG-16 network — all 13 convolutional layers and 5 max-pooling stages across Block 1 through Block 5 — has been simulated end-to-end and verified 100% functionally correct.

Verified in behavioural simulation (XSim), including:

- ✓ Full DDR3 read/write dataflow (weights + feature maps)
- ✓ Kernel-loading handshake sequencing between layers
- ✓ Inline BRAM-to-BRAM channel accumulation
- ✓ Max-pooling trigger and completion across all pooling stages
- ✓ Bit-accurate correctness validated against a Python/OpenCV golden model — zero overflow or saturation errors across the entire pipeline

1 Core Architecture

1.1 4-Channel Parallel CNN Engine

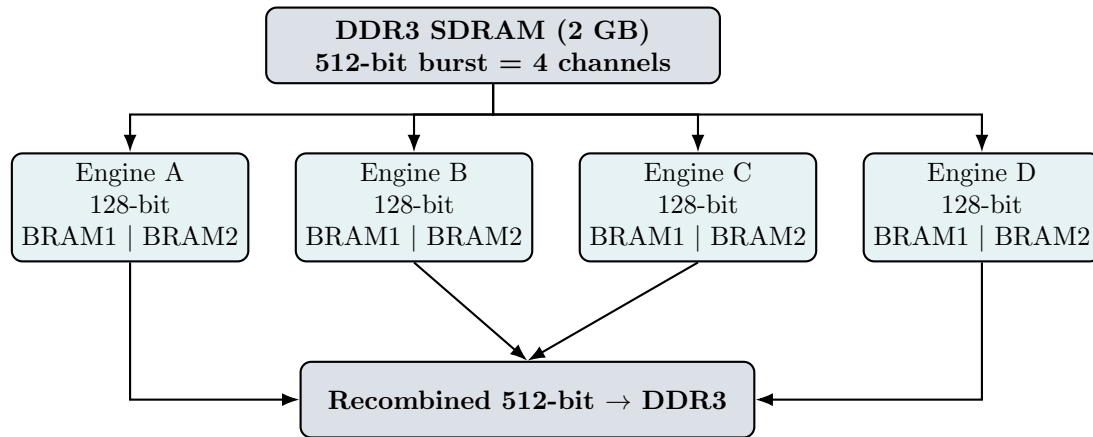
The accelerator instantiates **4 independent CNN core engines (A, B, C, D)**, each operating on a dedicated **128-bit** datapath in parallel — sustaining $72 \times 4 = 288$ concurrent MAC operations per clock cycle.

Each CNN core contains **two True Dual-Port BRAMs**:

BRAM2 supports **both standard convolution and depthwise convolution** — once all channels for a given filter are accumulated, activation is applied with zero additional pipeline overhead.

BRAM	Role	Port A	Port B
BRAM1	Input image / feature-map buffer	Fed directly from DDR3	Streams data continuously into the inline sliding-window convolution buffer
BRAM2	Output channel accumulation buffer	Read-modify-write accumulation	Bias + ReLU applied inline on final channel pass

1.2 DDR3 ↔ 4-Engine Data Split



- DDR3 delivers a **512-bit packet** per burst, carrying 4 channels' worth of data
- The packet is split into $4 \times 128\text{-bit}$ segments, one per engine, via BRAM1 Port A
- Once 4 filter passes complete, each engine's 128-bit result is **recombined into a 512-bit packet** and written back to DDR3
- The same split/combine mechanism governs max-pooling output

1.3 Data Precision — Weight vs. Pixel

Data Type	Bit-width	Purpose
Kernel Weight	8-bit	Compact on-chip / DDR3 storage
Image / Feature-map Pixel	16-bit	Preserves activation precision

This asymmetric precision strikes a deliberate balance between model accuracy and on-chip BRAM/DDR3 storage efficiency.

2 Worked Throughput Example

Input: $224 \times 224 \times 3$ image, 8 output filters → **Output:** $224 \times 224 \times 8$

$$\begin{aligned}
 \text{Required clock cycles} &= \frac{224 \times 224 \times 3 \times 8}{8 \times 4} \\
 &= \frac{1,204,224}{32} \\
 &= 37,632 \text{ cycles}
 \end{aligned}$$

At a clock period of 5 ns (200 MHz):

$$\begin{aligned}\text{Total time} &= 37,632 \times 5 \text{ ns} \\ &= 188.16 \mu\text{s}\end{aligned}$$

This formula generalizes to any image size, channel count, and filter count — and scales proportionally across all VGG-16 layers from Block 1 through Block 5.

3 Flexibility — Beyond VGG-16

The core CNN engine is **fully parameterized**, not hard-wired to VGG-16 alone. With only configuration-register changes — image size, channel count, filter count, kernel-pick counts, and DDR3 burst timing — the same architecture readily supports:

Network	Status
VGG-16	✓ Fully implemented & 100% verified
MobileNet-V2	✓ Supported (depthwise convolution built-in)
YOLOv3	✓ Supported
YOLOv3-tiny	✓ Supported
ResNet-18 / ResNet-50 / GoogleNet / Inception-V4	Requires 5×5 & 7×7 kernel extension in CNN core

Currently supported features:

- ✓ Kernel size: 3×3 (extendable to larger kernels)
- ✓ Stride: 1, 2, 3...
- ✓ Image size: fully parametrized (rows arbitrary, columns divisible by 8)
- ✓ Max-pooling
- ✓ Batch Normalization + ReLU
- ✓ Standard **and** Depthwise convolution, per channel/filter

To additionally support ResNet-18/50, GoogleNet, and Inception-V4, only the **CNN core engine** requires extension to add 5×5 and 7×7 kernel support (not yet implemented). The rest of the architecture — Block Controller, DDR3 interface, and hierarchical FSMs — remains **unchanged**.

4 Key Performance Results (VGG-16, Full Network)

7.7× fewer DSPs than comparable Virtex-7 VC707 designs in prior literature, while sustaining competitive throughput and closing timing at 200 MHz with zero failing endpoints.

5 Comparison with State-of-the-Art

On the identical Virtex-7 VC707 platform, Yuan et al. achieves lower latency but requires 7.7× more DSPs and 2.2× more LUTs — this design optimizes for area efficiency and resource-constrained deployment rather than raw peak speed.

Metric	Value
Operating Frequency	200 MHz (5 ns period)
Total VGG-16 MACs	15.5 GMAC per image
Sustained Throughput	41.9 GMAC/s (83.8 GOP/s)
End-to-End Latency	369.67 ms per image
Parallel CNN Engines	4 (Engine A/B/C/D)
LUT Utilization	32.52% (98,735 / 303,600)
BRAM Utilization	37.67% (388 / 1,030)
DSP Utilization	10.71% (300 / 2,800)
FF Utilization	12.74% (77,375 / 607,200)
Total On-Chip Power	4.93 W
DDR3 Overhead	45.67 ms ($\approx 12.4\%$ of total latency)
Timing Closure	WNS +0.063 ns, 0 failing endpoints (71,424 paths)

Work	Platform	DSPs	LUTs	Freq (MHz)	Latency (ms)	Throughput
Mei et al.	Virtex VX690T	1,728	210,992	200	151.9	202.4 GOP/s
Li et al.	Virtex VX690T	2,160	—	150	106.6	290.0 GOP/s
Yuan et al.	Virtex-7 VC707	2,296	215,556	—	29.6	—
Zhang & Zhang	Zynq-7020	138	—	130	182.3	41.2 GOP/s
This Work	Virtex-7 VC707	300	98,735	200	369.7	83.8 GOP/s

6 Repository Contents

This repository includes:

- ✓ Complete Verilog RTL source (Block Controller, 4× CNN Engines, DDR3 interface, FSM control)
- ✓ Thesis Paper
- ✓ IEEE-format Research Paper
- ✓ Project Presentation (PPT)

7 Full Project Archive

Due to GitHub's file size limits, the full project archive (~ 500 MB — including complete Vivado project files, testbenches, simulation waveforms, and build scripts) is hosted externally:

Download Full Project Archive (Google Drive):

<https://drive.google.com/file/d/1-X0kdnOKZGbk76jt35BEpLkjwY0c8G-/view?usp=sharing>

Contents: Full RTL source, DDR3 interface modules, testbenches, Vivado project (.xpr), and simulation build scripts for the FPGA-based VGG-16 CNN accelerator (Xilinx Virtex-7 VC707).

8 Tools & Platform

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Component	Details
HDL	Verilog
Target FPGA	Xilinx Virtex-7 VC707 (xc7vx485tffg1761-2)
Toolchain	Xilinx Vivado 2016.4 (licensed)
Simulator	Vivado XSim (behavioural DDR3 model)
Verification	Python / OpenCV golden-model bit-accurate cross-checking
Memory	DDR3 SDRAM (2 GB) via MIG 7-Series, 512-bit AXI4

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